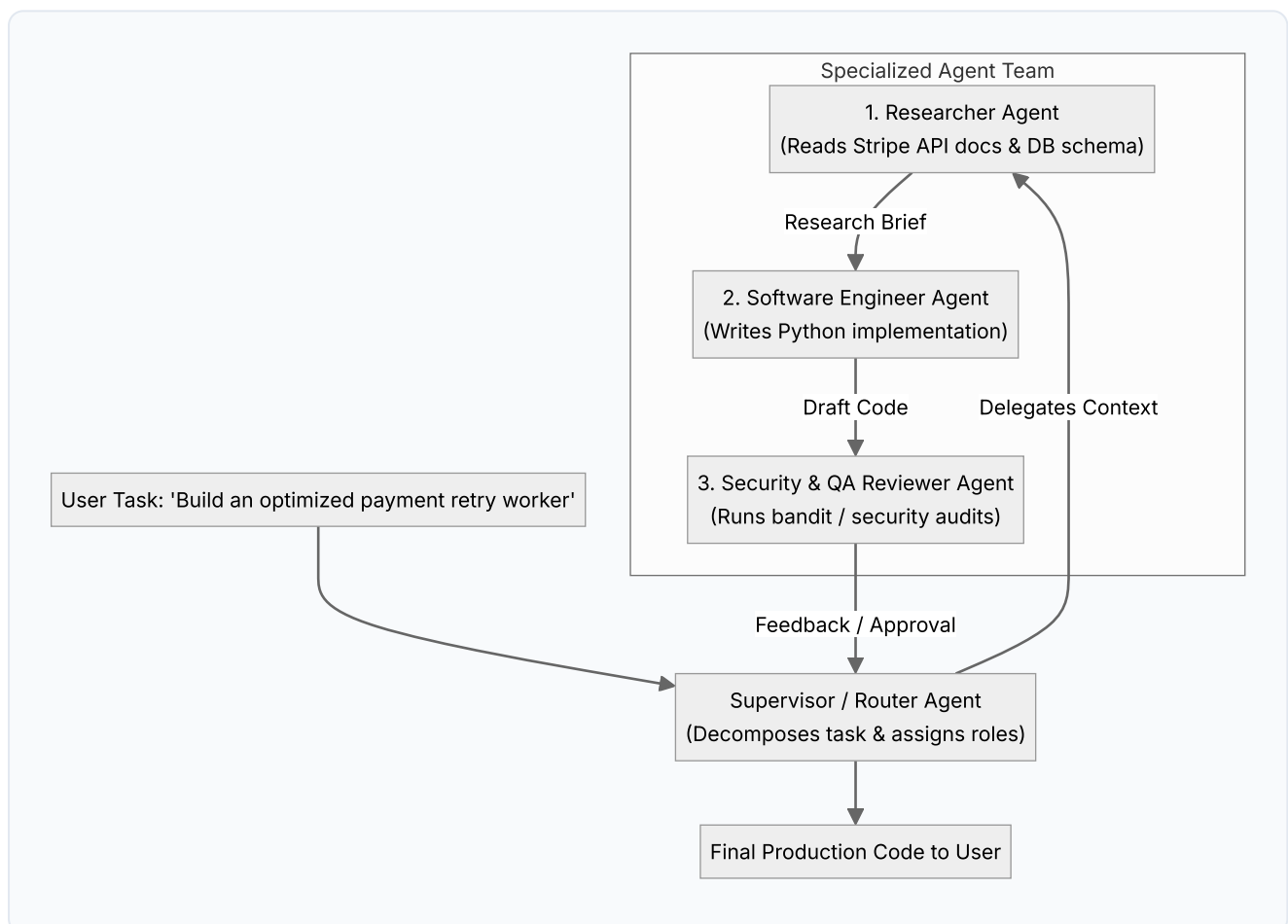


Agentic AI Chapter 7: Multi-Agent Collaboration & Production Deployment

Core Learning Objective: Master planetary-scale multi-agent systems. Build Supervisor-Worker swarms, implement peer-to-peer handoffs, stream token deltas via Server-Sent Events (SSE), and configure distributed tracing with OpenTelemetry and LangSmith.

1. Multi-Agent Topologies

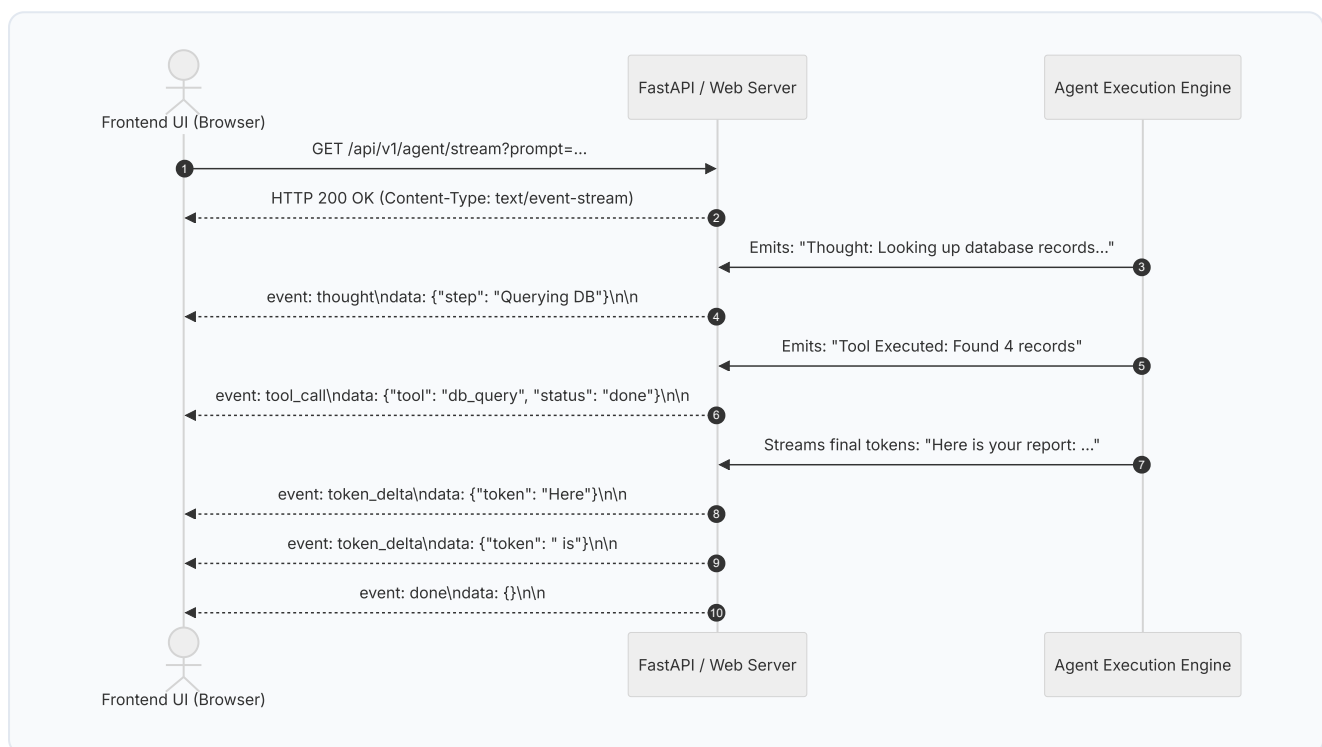
As tasks grow complex, a single monolithic agent with 30 tools fails due to context pollution and tool selection confusion. The solution is **Specialized Multi-Agent Teams**:



2. Multi-Agent Supervisor (Simulation: Scripted Agents, No Real LLM)

3. Real-Time Streaming Architecture (Server-Sent Events - SSE)

Users will not wait 30 seconds staring at a blank screen while an agent reasons and invokes tools. You must stream intermediate thoughts and token deltas in real-time using **Server-Sent Events (SSE)**:



4. Production Observability: OpenTelemetry & LangSmith

Every production agent deployment must log comprehensive execution spans:

- Trace ID:** Uniquely tracks a user request across all subagents and tools.
- Span Metrics:**
 - Exact token consumption (Prompt Tokens + Completion Tokens).
 - Wall-clock latency per LLM call vs per tool execution.
 - Model temperature, system prompt version, and exact JSON payloads.
- Error Traces:** Stack traces for flaky external tool APIs.

What the Simulation Above Does and Does Not Show

The supervisor example uses scripted agents (fixed strings) so that the control flow is deterministic and testable. It demonstrates delegation, result passing and a final merge, **not** real model behaviour. For a verified graph with checkpointing and approval see [examples/ex02_langgraph_hitl.py](#); to measure whether a multi-agent design beats a single agent, use the repeated-trial harness in [examples/ex05_eval_harness.py](#): report `pass^k` and a Wilson interval, not a single run.

Cost reality check. Multi-agent runs multiply tokens (reported around 15x a chat for research agents versus about 4x for a single agent). If the harness shows no statistically clear gain over the single-agent baseline, ship the single agent.

Further Reading

- [Anthropic: how we built our multi-agent research system](#)
- [AutoGen documentation](#)
- [OpenTelemetry documentation](#)

Check Yourself

Answer in your head or on paper first, then open each answer.

- ▶ 1. When is a single agent better than multi-agent?
- ▶ 2. What is a handoff?
- ▶ 3. Why bound the number of agents and rounds?
- ▶ 4. How do you decide if multi-agent helped?